

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

1. **An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.**

Use **Negative Transformation**. It reverses the gray levels of the image, converting bright regions into darker ones and vice versa. This reduces excessive brightness and improves visibility of details in bright areas.

2. **A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.**

Use **Log Transformation**. It expands low-intensity pixel values and compresses high-intensity values. This makes dark regions brighter and reveals hidden details.

3. **An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.**

Use **Histogram Equalization**. It redistributes the intensity values over the full gray-level range. This improves contrast and makes image details more visible.

4. **A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.**

Apply **Histogram Equalization**. It spreads pixel intensities uniformly across the histogram, increasing contrast and improving overall image quality.

5. **An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.**

Use **Adaptive Histogram Equalization (CLAHE)**. It enhances contrast in small regions instead of the whole image. This corrects non-uniform illumination without over-enhancing noise.

6. **A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.**

Use the **Mean (Average) Filter**. It replaces each pixel with the average of its neighboring pixels. This smooths random noise but may slightly blur edges.

7. **An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.**

Use the **Median Filter**. It replaces each pixel with the median value of neighboring pixels. It effectively removes noise while preserving image edges.

8. **An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.**

Use the **Laplacian Filter**. It is a sharpening filter that enhances high-frequency components such as edges. This restores image sharpness and fine details.

9. **Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.**

Use **High-Pass Filtering**. It emphasizes high-frequency components like edges and textures while suppressing smooth regions. This improves detail visibility.

10. **A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.**

Apply the **Median Filter** before further processing. It removes impulse noise effectively while maintaining edge information, making later analysis more accurate.

11. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

Use a **Low-Pass Filter** (Gaussian, Butterworth, or Ideal). It removes high-frequency noise while preserving important low-frequency image information.

12. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

Apply a **Gaussian Low-Pass Filter**. It smooths the image by removing high-frequency noise while keeping important low-frequency structures intact.

13. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

Use a **High-Pass Filter**. It removes low-frequency components and enhances edges and fine image details by preserving high-frequency information.

14. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

Use a **Gaussian Low-Pass Filter** in the frequency domain. It suppresses high-frequency components and produces a smooth, less noisy image.

15. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Use **Threshold Segmentation**. Pixels with intensity greater than the threshold are classified as the object, while the remaining pixels become the background.

16. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Apply **Global Thresholding**. A single threshold value separates foreground from background, making object extraction simple and effective.

17. An image requires identification of object edges.

Explain and justify a suitable edge detection method, describing its working and effectiveness.

Use the **Canny Edge Detector**. It detects edges accurately using gradient information and suppresses noise, producing thin and continuous edges.

18. Adjacent pixels have similar intensity values.

Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

Use **Region Growing**. Starting from selected seed points, neighboring pixels with similar intensity are grouped together to form meaningful regions.

19. Detected edges in an image appear broken.

Explain and justify a method to improve edge continuity, including its role in post-processing.

Use **Region Growing**. Starting from selected seed points, neighboring pixels with similar intensity are grouped together to form meaningful regions.

20. You need to extract object outlines from an image.

Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Use **Boundary Extraction**. The boundary is obtained by subtracting the eroded image from the original image. This clearly highlights object outlines.

21. An image suffers from excessive brightness variation between regions.

Explain and justify a suitable histogram processing technique to normalize intensity distribution, describing its working and expected outcome.

Use **Histogram Equalization**. It normalizes the intensity distribution across the image, reducing brightness variations and improving contrast.

22. A medical image requires enhancement of subtle intensity differences. Describe and justify a suitable grey level transformation to highlight fine variations, explaining its effect on diagnostic visibility.

Use **Contrast Stretching**. It expands the intensity range, making subtle differences in tissues more visible and improving diagnostic accuracy.

23. An image contains salt-and-pepper noise. Explain and justify the most appropriate spatial filter, describing its working mechanism and effectiveness in noise removal.

Use the **Median Filter**. It replaces noisy pixels with the median of neighboring pixels, effectively removing impulse noise while preserving edges.

24. An image appears overly smooth after filtering, losing important features. Explain and justify a sharpening technique to restore details, including its working and impact on edges.

Use **Unsharp Masking**. It sharpens the image by adding edge information back to the original image, restoring lost details.

25. An image contains both low-frequency background and high-frequency noise. Describe and justify a filtering approach that preserves important structures while reducing noise, explaining the concept clearly.

Use a **Gaussian Low-Pass Filter**. It reduces high-frequency noise while preserving important image structures and smooth background variations.

26. An image requires enhancement of edges without amplifying noise significantly. Explain and justify a suitable spatial or frequency domain method, describing its operation and expected result.

Use **High-Boost Filtering**. It sharpens edges by enhancing high-frequency components while retaining the original image, reducing excessive noise amplification.

27. A grayscale image shows poor separation between object and background intensities.

Use **Otsu's Thresholding**. It automatically selects the optimal threshold by minimizing intra-class variance, resulting in accurate segmentation.

Describe and justify a thresholding approach that improves segmentation accuracy, explaining its working.

28. An image contains multiple regions with similar intensity but different textures.

Explain and justify a region-based segmentation approach, describing how it groups pixels effectively.

Use **Region-Based Segmentation (Region Growing)**. It groups connected pixels based on intensity similarity and spatial relationship, effectively separating textured regions.

29. Detected edges are noisy and fragmented.

Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.

Apply **Morphological Closing** after edge detection. It removes small gaps, connects fragmented edges, and improves boundary detection.

30. An application requires precise object boundary extraction for shape analysis.

Explain and justify a suitable segmentation and boundary detection approach, describing its working and effectiveness.

Use **Canny Edge Detection followed by Boundary Extraction**. Canny detects accurate edges with low noise, and boundary extraction produces clear object outlines for shape analysis.

Code of Conduct:

*I **KANMANI K B (192524306)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Rubrics for Evaluation

<i>Criteria</i>	<i>Weight age</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>

		<i>required points</i>			
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understanda ble with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>